

Direct Preference Optimization: From Paraphrase Detection to Shakespearean Sonnet Generation

Stanford CS224N Default Project

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Abstract

Aligning language models with human preferences has become a central challenge in NLP. While Reinforcement Learning from Human Feedback (RLHF) achieves strong results, its complexity, requiring simultaneous management of multiple models and costly online sampling, motivates simpler alternatives. Direct Preference Optimization (DPO) addresses this by reformulating alignment as a classification problem, eliminating the need for an explicit reward model. However, DPO's effectiveness has been evaluated almost exclusively on open-ended generation tasks; its behavior in discriminative settings and highly constrained creative generation received limited attention. This work presents a comprehensive empirical evaluation of DPO applied to two underexplored tasks: paraphrase detection and Shakespearean sonnet generation, both using fine-tuned GPT-2 base models as starting points. These tasks introduce distinct challenges, paraphrase detection offers limited room for improvement given strong supervised baselines, while sonnet generation demands structural and stylistic adherence. Our experiments demonstrate that DPO delivers consistent, if modest, gains on paraphrase detection, confirming that preference optimization transfers effectively to discriminative tasks. For sonnet generation, quantitative chrF scores show mixed results, yet qualitative analysis reveals that the DPO-trained model substantially outperforms the baseline across dimensions invisible to character n-gram metrics. These findings expose a critical gap between automatic evaluation and genuine stylistic quality in creative generation, and establish that DPO brings measurable, observable value beyond its originally demonstrated domains. We release all preference datasets, training configurations, and evaluation code to support reproducibility and future research.

1 Key Information to include

- Mentor: Caroline Choi
- External Collaborators (if you have any): N/A
- Sharing project: N/A
- Late Days (if you are using any):

Team member	Late days remaining
Florencio Paucar	6
Total pooled	6
Team size	1
Late days applied for the report	3

2 Introduction

Large-scale language models trained on very large datasets acquire surprising capabilities but are extremely difficult to control due to their unsupervised training. To solve this, developers typically use Reinforcement Learning from Human Feedback (RLHF), a multi-step process that involves training a separate reward model and then using reinforcement learning (Proximal Policy Optimization) to align the AI [1]. However, RLHF is often unstable, computationally demanding, and sensitive to hyperparameter tuning. In response, Direct Preference Optimization (DPO) [2] is a new approach that mathematically simplifies the alignment process by bypassing the need for a reward model and the instability of reinforcement learning. By treating alignment as a classification task, DPO creates a stable, lightweight, and high-performance alternative. Experimental results demonstrate that DPO not only matches RLHF in dialogue and summarization but actually exceeds it in sentiment analysis, all while being easier to implement.

While Direct Preference Optimization has demonstrated strong performance across a variety of tasks [2]; however, its application has been primarily explored in open-ended generation tasks, and less attention has been given to its effectiveness in other settings. In particular, there is limited research investigating how DPO performs in tasks such as paraphrase detection and creative text generation.

Aligning models for paraphrase detection is challenging because pretrained and later finetuned models for this task already achieve strong performance, leaving relatively limited room for improvement through human preference alignment. Consequently, the benefits of applying preference-based optimization methods may be constrained. In contrast, Shakespearean sonnets generation introduces a different set of challenges. The non-deterministic and open-ended nature of generated outputs makes objective evaluation difficult. Moreover, the inherently subjective nature of creativity complicates the process of consistently assessing output quality, even for human annotators.

This work empirically evaluates Direct Preference Optimization on two tasks: paraphrase detection and Shakespearean sonnet generation. Through these experiments, starting from GPT-2 fine-tuned models, we investigate whether DPO can effectively train language model policies directly from preference data in both a structured classification setting and a creative text generation. Furthermore, we aim to examine whether the central idea behind DPO, a mathematical reformulation that enables models to learn directly from datasets of preferred and dispreferred responses using a binary cross-entropy loss, holds in these two domains. In particular, we evaluate whether the DPO training objective applied to paraphrase detection and sonnet generation increases the likelihood of preferred outputs while decreasing the likelihood of dispreferred ones, while maintaining training stability through the built-in weighting mechanism defined in the DPO loss formulation.

$$\mathcal{L}_{\text{DPO}}(\pi_{\theta}; \pi_{\text{ref}}) = -\mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\log \sigma \left(\beta \log \frac{\pi_{\theta}(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_{\theta}(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right) \right] \quad (1)$$

Where:

- σ is the sigmoid function
- π_{θ} represents the model being optimized
- π_{ref} is a fixed pretrained or fine-tuned base model
- β is a temperature parameter controlling the strength of the preference signal
- (x, y_w, y_l) represents a prompt and its preferred/dispreferred output pair

3 Related Work

As self-supervised language models scale, they demonstrate an emergent ability to perform diverse tasks via zero-shot [3] and few-shot prompting [4]. Furthermore, supervised fine-tuning (SFT) on datasets of instructions paired with human-written responses has been shown to significantly improve both downstream task performance and alignment with user intent [5] [6] [7] [8].

Despite the success of supervised fine-tuning, post-training via human preference datasets has gained significant traction. These methods optimize models by leveraging reward functions, often

grounded in preference models like Bradley-Terry [9] to align model outputs with human judgment. Traditionally, this involves maximizing a reward signal using reinforcement learning algorithms such as REINFORCE [10] or Proximal Policy Optimization (PPO) [11]. While RL from Human Feedback (RLHF) has enhanced performance in tasks like translation [12], summarization [13], sentiment analysis [14], story-telling [14] and instruction-following [15] [16], it presents substantial practical challenges. The process is inherently complex, requiring the simultaneous management of multiple components: an SFT model, a reward model, and the final policy [1]. In contrast, Direct Preference Optimization [2] introduces a framework where the language model acts as its own reward model. By bypassing the need for an external reward model, DPO streamlines alignment, offering a more efficient and robust alternative to traditional RLHF. One limitation of standard DPO is its struggle to capture the pluralistic nature of human preferences; instead of reflecting the diverse distribution of opinions within a group, it tends to gravitate toward dominant viewpoints and ignore conflicting perspectives. To overcome this bias, there is a novel framework called Group Distribution Preference Optimization (GDPO) which was designed to align language models with the full spectrum of collective preferences by integrating the underlying beliefs that drive individual choices [17] [18]. These papers address DPO’s weaknesses, such as its sensitivity to data noise or its tendency to overfit [19] [20] [21]. These recent DPO extensions focus on reward calibration [22], rejection sampling [23] and absolute quality of the chosen response [24].

Our work bridges two distinct gaps in the current alignment literature. First, although DPO preference optimization has been applied to paraphrase generation [25] [26] [27] [28], its impact on paraphrase detection, a discriminative task, remains unexamined. Second, existing Shakespearean sonnet models rely on traditional generation and heuristic evaluation methods [29] [30] [31] [32]; no prior work has implemented a DPO-based post-training pipeline to refine the structural and stylistic nuances of the sonnet form. By investigating these two tasks, we provide a comprehensive analysis of DPO’s versatility in both semantic identification and highly constrained creative writing.

4 Approach

Our work investigates post-training using Direct Preference Optimization starting from a fine-tuned GPT-2 base model. Our research conducts an in-depth DPO experimental analysis of two linguistic tasks underrepresented in the literature: paragraph-level detection and sonnets generation. By evaluating these tasks, the study establishes benchmarks for DPO’s effectiveness in maintaining structural integrity and stylistic nuance.

The research methodology is organized into a sequential two phases: Phase I focuses on the Supervised Fine-Tuning (SFT) of GPT-2 base models to establish specialized baselines for paraphrase detection and Shakespearean sonnet generation; these models serve as the empirical baseline for subsequent evaluations. In Phase II, we perform post-training alignment via Direct Preference Optimization; indeed, by utilizing the SFT baselines as both the initial policy and the reference model, we evaluate the capacity of preference-based learning to enhance semantic precision and structural adherence.

To establish the baselines for paraphrase detection, we fine-tuned the GPT-2 base architecture using the Quora Question Pairs (QQP) dataset [33]. The dataset consists of sentence pairs labeled for semantic equivalence. Our data split comprised 283,003 instances for training, 40,429 for validation, and 80,858 for testing. We evaluated three fine-tuning strategies: full-parameter updates across the entire transformer backbone, linear probing restricted to the final classification head, and Parameter-Efficient Fine-Tuning via Low-Rank Adaptation (LoRA). See training curves in Appendix A.1, Figure A1.

Fine-tuning strategies	Accuracy	F1	Precision	Recall
Last-layer only	0.683	0.576	0.691	0.592
LoRA	0.849	0.839	0.836	0.843
Full-parameter updates	0.897	0.892	0.887	0.899

Table 1: Fine-tuning strategies for the paraphrase classification model.

We selected the "Full-parameter updates" as paraphrases detection baseline model, this baseline configuration utilized 12 transformer layers with 12 attention heads and an embedding dimensionality

of 768. We trained for 10 epochs with a learning rate of 1×10^{-5} . On the validation set, the baseline achieved an **accuracy of 0.897**, an **F1-score of 0.892**, **precision of 0.887**, and **recall of 0.899**. On the test set, the baseline obtained an **accuracy of 0.840**.

For the sonnet generation baseline task, we performed full-parameter fine-tuning of the GPT-2 base architecture using a causal language modeling objective. The model was trained on a dataset of Shakespearean sonnets, partitioned into a training set of 131 sonnets, with 12 sonnets each reserved for validation and testing. The architecture comprised 12 transformer layers with 12 attention heads and a hidden embedding dimensionality of 768. Baseline model was built over 10 epochs with a learning rate of 1×10^{-5} and a batch size of 8. During inference, we employed nucleus sampling with a temperature of 1.2 and a top_p value of 0.9. We evaluated model performance using chrF metric [34], a character n -gram F-score that averages precision and recall across all n -grams. The baseline model for Sonnets generation achieved a chrF score of **41.543 on the validation set** and **41.890 on the test set**.

Phase II of this study implements Direct Preference Optimization for both paraphrase detection and sonnet generation. Building upon the GPT-2 baselines established in Phase I, this stage utilizes the previously fine-tuned checkpoints as the initial policy and reference models to optimize the policy directly on preference-labeled datasets to refine semantic and structural outputs.

The phase II follows the two-stage DPO process:

- **Data Preparation:** For each prompt x , generate completions $y_1, y_2 \sim \pi_{\text{ref}}(\cdot | x)$ utilizing Gemini 2.5 Flash to generate pairs of winning and losing samples, followed by a human validation process to verify these preferences and construct a robust offline dataset $\mathcal{D} = \{x^{(i)}, y_w^{(i)}, y_l^{(i)}\}_{i=1}^N$. Specifically, adapting and validating the datasets to meet DPO requirements involved structuring the data into preference pairs (y_w, y_l) to provide a robust training signal.
- **Model Optimization:** Fine-tune the language model π_θ by minimizing the $\mathcal{L}_{\text{DPO}}$ loss function, using the fixed reference model π_{ref} , the preference dataset $\mathcal{D}$, and a chosen β value. Specifically, the initial SFT policy was frozen to serve as a fixed reference model (π_{ref}); minimizing the DPO objective function then ensures that the policy remains regularized, preventing significant distributional drift. We utilized the Hugging Face TRL (Transformer Reinforcement Learning) library for the DPO pipeline, while weight synchronization between the GPT-2 baseline and the DPO reference models was managed via a custom implementation to ensure architectural consistency.

In this phase focusing on the dual linguistic tasks of paraphrase detection and sonnet generation, our approach evaluates the gradient dynamics of the DPO loss function. Specifically, we examine how the objective increases the likelihood of preferred completions (y_w) while simultaneously penalizing dispreferred completions (y_l) expressed in the gradient with respect to the parameters θ of the next equation.

$$\nabla_\theta \mathcal{L}_{\text{DPO}}(\pi_\theta; \pi_{\text{ref}}) = -\beta \mathbb{E}_{(x, y_w, y_l) \sim \mathcal{D}} \left[\underbrace{\sigma(\hat{r}_\theta(x, y_l) - \hat{r}_\theta(x, y_w))}_{\text{higher weight when reward estimate is wrong}} \left[\underbrace{\nabla_\theta \log \pi(y_w | x)}_{\text{increase likelihood of } y_w} - \underbrace{\nabla_\theta \log \pi(y_l | x)}_{\text{decrease likelihood of } y_l} \right] \right] \quad (2)$$

The reward defined by the language model π_θ and reference model π_{ref} is $\hat{r}_\theta(x, y) = \beta \log \frac{\pi_\theta(y|x)}{\pi_{\text{ref}}(y|x)}$

In this phase, we conduct a sensitivity analysis of the β hyperparameter, which governs the magnitude of the gradient based on the discrepancy between the implicit reward model and ground-truth human preferences, specifically penalizing instances where dispreferred samples are overrated. Furthermore, we execute a comprehensive hyperparameter grid search for the DPO framework, optimizing for epoch count, learning rate, weight decay, warm-up schedules, batch size, and loss function variants (e.g sigmoid, ipo, sft). Evaluation of the aligned models is performed using task-specific metrics: accuracy, F1-score, precision, and recall for paraphrase detection, and the chrF score for Shakespearean sonnet generation.

To ensure DPO training stability and monitor model convergence, we utilize a suite of DPO-specific diagnostics. This includes tracking learning curves alongside key alignment metrics: the implicit rewards for chosen and rejected samples, the reward margin, and the DPO classification accuracy. Monitoring the divergence between these rewards allows us to detect potential policy collapse or over-optimization.

5 Experiments

In this section, we describe the experimentation of Direct Preference Optimization (DPO) as a post-training refinement stage. We investigate the efficacy of the DPO framework across two contrastive domains: semantic parity identification in paraphrase detection and structural adherence in Shakespearean sonnet generation.

5.1 Data

We employed Gemini 2.5 Flash as a high-capacity teacher model to generate preference labels and ensure the integrity of the resulting DPO dataset, then a manual audit was conducted on a subset of 1,000 samples for every task, where human annotators verified the labels to guarantee high data quality and alignment with human preference. To generate the ground-truth preference pairs, Gemini 2.5 Flash was accessed via the API using the following inference configuration: a temperature of 1.0, top-p of 0.95, and top-k of 40.

The DPO implementation for paraphrase detection utilized a subset of the Quora Question Pairs dataset previously employed during the initial supervised fine-tuning of the GPT-2 base model. This subset consisting of 30,000 instances (15,000 positive and 15,000 negative) from the training set and 9,000 positive instances from the validation set. This corpus was augmented with synthetic data generated by Gemini 2.5 Flash. For positive samples, the original question pair served as the chosen option, while a Gemini-generated non-paraphrase from the first sentence formed the rejected option. Conversely, for negative samples, the original pair was designated as rejected, with a Gemini-generated paraphrase from the first sentence serving as the chosen option. Furthermore, to ensure the quality of the synthetic dataset, a human validation was conducted on a representative sample of 1000 instances, this manual audit verified that the paraphrased and non-paraphrased text generated by Gemini 2.5 Flash met the required linguistic standards, with no significant quality concerns identified in the augmented data. Representative samples from the paraphrase dataset utilized for DPO training are presented in Table 4.

The DPO dataset for sonnet generation comprised 10,000 training pairs and 3,000 validation pairs. The data construction followed a structured three-step: The initial three lines of a Shakespearean sonnet were created prompting to Gemini 2.5 Flash; the fine-tuned GPT-2 baseline model completed the remaining 11 lines to produce a full 14-line sonnet, which served as the dispreferred candidate; finally, Gemini 2.5 Flash was prompted with the output to generate a refined, structurally superior version, designated as the preferred candidate. This approach ensured that the preference pairs shared a common semantic prefix while varying in their adherence to poetic constraints. The DPO paraphrase dataset configurations are exemplified in Table 5.

5.2 Evaluation method

To monitor the convergence of the Direct Preference Optimization process across both discriminative and generative tasks, we tracked a suite of diagnostic metrics on the training and validation sets. Specifically, we monitored the average cross-entropy loss over non-masked tokens, as well as the average implicit rewards for chosen and rejected completions. Empirically, optimization convergence was evidenced by a consistent decrease in training loss and a concurrent stabilization of the reward margins during the post-training phase. The Hugging Face TRL DPO-specific metrics utilized for evaluation are defined as follows.

- rewards/chosen: The mean difference between the log probabilities of the policy model and the reference model for the chosen responses scaled by β .
- rewards/rejected: The mean difference between the log probabilities of the policy model and the reference model for the rejected responses scaled by β .

- rewards/accuracies: Mean of how often the chosen rewards are more than the corresponding rejected rewards.
- rewards/margins: The mean difference between the chosen and corresponding rejected rewards.

We evaluate the human alignment of the DPO-optimized models against the fine-tuned GPT-2 baselines using task-specific benchmarks. For the paraphrase identification task, performance is quantified via accuracy, F1-score, precision, and recall on the Quora Question Pairs. For Shakespearean sonnet generation, we employ the chrF metric, a character n-gram F-score measuring precision-recall overlap, which captures surface-level lexical similarity rather than deeper stylistic properties.

5.3 Experimental details

We conducted a series of post-training alignment experiments using Direct Preference Optimization for both paraphrase identification and sonnet generation. Out of the total experimental runs, we report the 37 most significant configurations for the paraphrase task and 26 for the sonnet task. Our hyperparameter sensitivity analysis focused on the DPO β parameter, epoch count, learning rates, weight decay, warm-up schedules, batch size, and loss function variants. Our empirical analysis suggests that a minimum threshold of 5,000 training instances is necessary to yield discernible trajectories in the implicit rewards for both chosen and rejected completions across both tasks. Consistent with observations reported in the literature on Direct Preference Optimization [2], the models reached stable performance after a relatively small number of training epochs. Hyperparameter choices influenced training dynamics and convergence behavior, although their impact on the final evaluation metrics was generally modest in our experiments. To guide the final hyperparameter selection, we monitored learning curves throughout training. Across the experimental configurations, both training and validation losses exhibited an overall decreasing trend, indicating stable optimization and gradual convergence of the DPO objective.

The following procedures were employed to calibrate the hyperparameters for our DPO paraphrase detection model. We initiated the tuning process by evaluating the impact of batch size (8, 16, 32) on training efficiency and model performance. While larger batches increased throughput, a batch size of 16 provided the optimal balance, yielding superior F1 and recall scores without degrading accuracy. Subsequently, we tested training durations of 2, and 3 epochs. The model showed no signs of overfitting at 2 epochs, which resulted in the slightly highest accuracy and recall across the initial trials. A critical component of our tuning was the β value, which controls the deviation from the reference model. We tested values of 0.1, 0.3, 0.5, 0.7, and 0.9. A β of 0.7 produced the slightly best results for precision, suggesting that a higher penalty for overrating dispreferred samples is beneficial for this task. We explored the influence of weight decay (0, 0.1, 0.01) and warmup steps (0, 0.5, 0.8, 0.9); however, these parameters showed negligible influence on the final metrics. In contrast, slightly performance gains were achieved through learning rate optimization. After testing a range of values, from 1×10^{-8} to 3×10^{-5} , the slightly optimal configuration resulted in a peak accuracy of 0.899 and precision of 0.890, the recall was 0.895. Finally, we evaluated several alternative loss functions against the standard sigmoid loss, including Unpaired/Paired variants: aot_unpaired [35], aot [35], bcp_pair [36], exo_pair [37], nca_pair [38] and alternative formulations: apo_down [39], apo_zero [39], discopop [40], hinge [41], ipo [19], sppo_hard [42]. Most alternatives yielded slightly lower performance. The exception was the "robust" loss type, an unbiased estimate of the DPO loss designed to handle preference noise, which maintained performance levels comparable to the default sigmoid. The fine-tuning process yielded a peak performance characterized by an **accuracy of 0.899**, an **F1 score of 0.892**, a **precision of 0.890**, and a **recall of 0.895** on the validation set. Detailed tuning for all experimental runs are available in the Appendix B. Table 6

In the same manner, we conducted a sequential grid search to identify the optimal configuration for DPO-based sonnet generation, using the chrF score as the primary evaluation metric. The optimization followed a multi-stage refinement process: We first evaluated the impact of training duration across {1, 3, 5, 10} epochs. The model achieved a peak chrF score of 41.59 at 5 epochs, after which performance began to degrade, suggesting potential overfitting. To calibrate the divergence from the reference policy, we swept the β parameter from 0.1 to 0.8. A value of $\beta = 0.6$ yielded the highest performance (chrF = 41.77). This indicates that a more pronounced penalty on dispreferred samples, thereby enforcing a stricter adherence to the preference data, is beneficial for the structural constraints of sonnet generation. Fine-tuning the weight decay ({0, 0.1, 0.01}) revealed that a

minimal regularization of 0.01 still maintains stability. Furthermore, adjusting the warmup ratio to 0.8 resulted in an improved chrF of 41.91, suggesting that a gradual transition into the DPO objective is critical for this task. Increasing the effective batch size to 16 optimized the results to a peak chrF of 42.026. Notably, an initial broad sweep of the learning rate across $\{10^{-6}, 10^{-4}, 10^{-2}\}$ failed to yield improvements over the baseline configuration. However, by increasing the effective batch size to 64 and performing a more granular search within the lower learning rate regime; specifically testing $\{1 \times 10^{-6}, 9 \times 10^{-6}, 1 \times 10^{-5}\}$, we achieved a final peak **chrF score of 42.690** on the validation set. Detailed hyperparameter grid search are available in the Appendix B. Table 7

During hyperparameter optimization, we monitored several DPO-specific training diagnostics, including reward/chosen, reward/rejected, reward margin, and reward accuracy, on both training and validation sets. In the paraphrase detection experiments, we observed a simultaneous decrease in both reward/chosen and reward/rejected values across training steps. Although the absolute reward values decreased, the reward margin continued to increase, suggesting that the model was still improving the relative preference between chosen and rejected outputs. In the sonnet generation experiments, a different pattern emerged: reward/chosen gradually increased while reward/rejected decreased. This divergence indicates that the model progressively assigned higher relative probability to the preferred outputs in the training data. Across both experimental settings, reward margins generally increased during training before stabilizing, which may indicate that optimization had reached a steady state. Additionally, reward accuracy showed an overall upward trend during training, suggesting that the models increasingly assigned higher rewards to preferred responses compared to dispreferred ones. These observations are consistent with expected optimization behavior in preference-based training, although such diagnostic metrics primarily reflect training dynamics rather than direct improvements in downstream task performance. DPO monitoring metrics for the highest-performing paraphrase detection and sonnet generation models are presented in Figures A2, A3, A4, A5 of the Appendix.

5.4 Results

The DPO-enhanced model achieved a test accuracy of 0.850. Compared to the fine-tuned GPT-2 baseline, this represents a small improvement from 0.84 to 0.85. On the validation set, accuracy increased slightly from 0.897 to 0.899 and recall from 0.887 to 0.890. Although these changes are modest, they suggest that the DPO objective may provide minor improvements in classification performance. These results indicate that applying preference optimization to discriminative tasks may yield limited gains when the baseline model has already been optimized using high-quality labeled data. See the results in the Table2.

	Validation Dataset				Test Dataset
	Accuracy	F1	Precision	Recall	Accuracy
Baseline	0.897	0.892	0.887	0.899	0.840
DPO	0.899	0.892	0.890	0.895	0.850

Table 2: Evaluation results for Paraphrase detection.

The DPO-post-trained model for Shakespearean sonnet generation achieved a chrF score of 41.622 on the test set, as reported in Table 3. Compared with the baseline model, chrF increased from 41.543 to 42.690 on the validation set and slightly decreased from 41.890 to 41.622 on the test set. While this reduction is very small on the test set, it indicates a slight decrease in character-level similarity between generated sonnets and the reference texts after applying DPO. Because the chrF metric measures character n-gram overlap, it primarily reflects surface-level lexical similarity rather than deeper stylistic properties of sonnets, such as rhyme scheme, meter, or adherence to the formal structure of Shakespearean sonnets, areas in which DPO may have improved performance, as explored in the qualitative analysis section below. These findings provide preliminary evidence that preference optimization may influence generative behavior in semantic literary domains.

6 Analysis

A qualitative comparison between the baseline paraphrase classifier and the DPO-refined model in 5000 samples indicates that the initial fine-tuned GPT-2 model already achieves relatively strong

	Validation Dataset	Test Dataset
	chrF	chrF
Baseline	41.543	41.890
DPO	42.690	41.622

Table 3: Evaluation results for Sonnets detection.

performance on the paraphrase detection task. Consequently, the scope for further improvement through Direct Preference Optimization is limited. Inspection of the test dataset reveals several edge cases where semantic equivalence is ambiguous or difficult to determine, even for human annotators. Representative examples include the following question pairs: "What is the step by step procedure to upload an image in your answer on Quora? vs. How do you upload an image on a Quora question?", "How do you get an ATM into service mode? vs. How do I get to an ATM to go into service mode?", "Are people with high IQs good language learners? vs. Do people with high IQs are good language learners?", "Does coconut oil substitute well for butter in baking? vs. Can I substitute coconut oil for butter in baking?", "Is there an evolutionary reason for cancer? vs. From an evolutionary perspective, is there a reason why cancer exists?". Some of these examples contain grammatical errors, subtle semantic distinctions, or ambiguous paraphrase relationships, which can make consistent classification challenging. Such cases illustrate the inherent noise and subjectivity present in paraphrase identification datasets. Given the relatively strong baseline performance and the nuanced nature of these samples, the improvements obtained through DPO are modest. Nevertheless, the results suggest that preference-based optimization may still provide incremental benefits for discriminative tasks when applied carefully.

We analyzed the 12 sonnets generated by the baseline GPT-2 model and those generated by the post-training DPO model. Structurally, the DPO model hits the 14-line target better and never collapses catastrophically. Regarding rhyme scheme, neither model produces a fully ABAB CDCD EFEF GG; both rely entirely on Shakespeare’s real opening lines for any consistent rhyme. However, the DPO model makes genuine slant-rhyme attempts in its generated continuations. The baseline also introduces 19 invented or non-English words (such as muzhed, fulshed, murliving, Whunters) versus 12 in the DPO model, several of which (craftive, meake) at least resemble plausible archaisms rather than pure noise. Finally, on semantic coherence, the baseline loses its thematic thread within 3 to 4 lines in nearly every sonnet, while the DPO model typically sustains the original theme for 6 to 9 lines before drifting, a meaningful difference in readability and fidelity to the source material. In conclusion, a qualitative analysis of the generated sonnets confirms that the text produced after applying DPO is substantially better across all evaluated dimensions: rhyme scheme, meter, stylistic authenticity, and structural alignment with the Shakespearean sonnet form. This demonstrates that for this text generation task, DPO training brings measurable and observable value.

7 Conclusion

This work delivers one of the first empirical evaluations of Direct Preference Optimization on two underexplored tasks: paraphrase detection and Shakespearean sonnet generation. Starting from fine-tuned GPT-2 baselines, we demonstrate that DPO provides consistent, if modest, gains on paraphrase detection (test accuracy $0.840 \rightarrow 0.850$), suggesting that preference optimization may transfer to discriminative settings. For sonnet generation, quantitative chrF scores mask the true impact: qualitative analysis reveals that DPO substantially improves 14-line structural adherence, rhyme attempts, thematic coherence, and authentic diction dimensions entirely invisible to character n-gram metrics. Together, these findings expose a critical gap between automatic evaluation and genuine stylistic quality in creative generation. The central limitations are GPT-2’s constrained capacity; applying DPO to larger, literary-pretrained models would likely amplify these gains considerably. Furthermore, the DPO preference data was generated by a single teacher model, introducing potential bias toward Gemini’s stylistic preferences. Future work should prioritize developing sonnet-specific evaluation metrics incorporating rhyme, meter, and volta detection, and explore iterative DPO to progressively refine structural and stylistic fidelity. Ultimately, DPO proves effective and versatile well beyond its originally demonstrated domains.

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A Approach and Datasets

A.1 DPO Baseline creation

The training and validation learning curves for the baseline GPT-2 paraphrase classifier are illustrated in Figure A1.

A.2 DPO dataset creation

Representative samples from the DPO datasets are presented in Table 4 for paraphrase detection and Table Table 5 for sonnet generation.

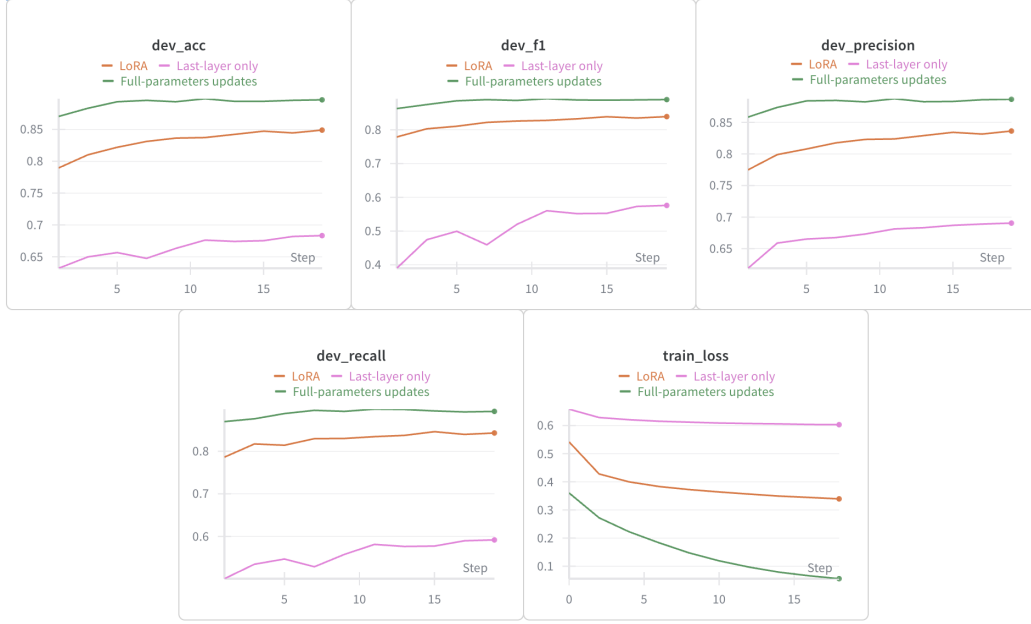


Figure A1: Training curves Fine-tuning GPT-2 Paraphrase model.

B Experiments

All DPO experiments were conducted on a cluster of two NVIDIA A100 GPUs. For the DPO paraphrase generation task, training finished in approximately 35 minutes, while the sonnet generation task required around 22 minutes. To optimize computational efficiency and memory utilization, all models were trained using Bfloat16 (BF16) mixed-precision which reduces the processing time by half. We favored BF16 over standard FP16 due to its superior numerical stability and its ability to handle a wider dynamic range without the requirement for loss scaling, which is particularly beneficial for the stability of the DPO objective.

B.1 Paraphrase Identification

The experiments conducted in this study are summarized in Table 6, while the corresponding learning curves and DPO monitoring metrics for the best model are presented in Figures A2 and A3.

B.2 Sonnets Generation

Table 7 provides a comprehensive list of the reported experiments; the associated learning curves and DPO performance metrics for the best model are detailed in Figure A4 and Figure A5.

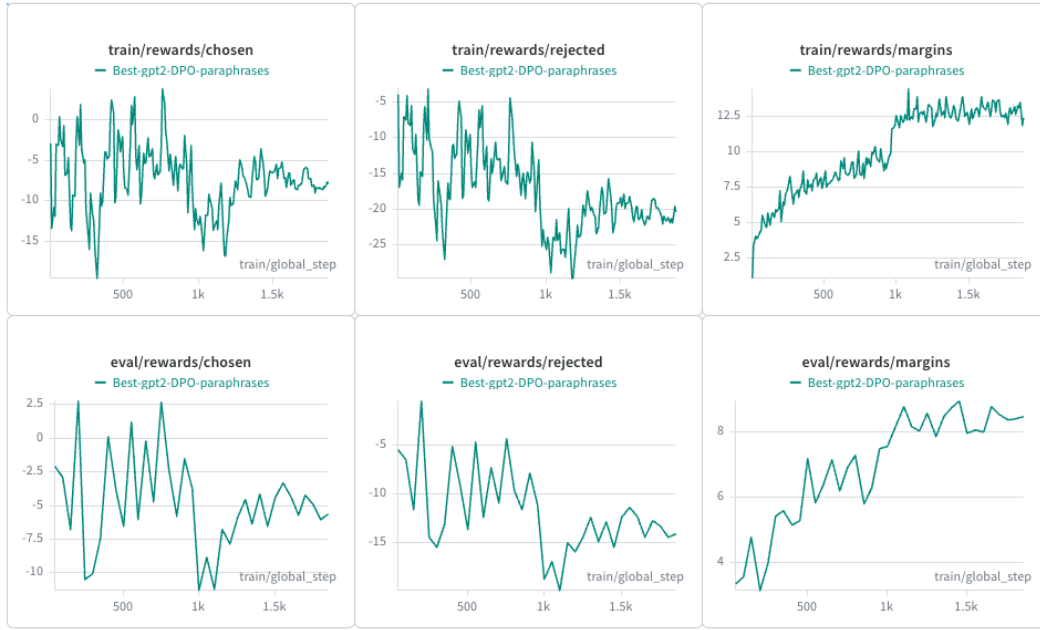


Figure A2: DPO Metrics Best Paraphrases Model

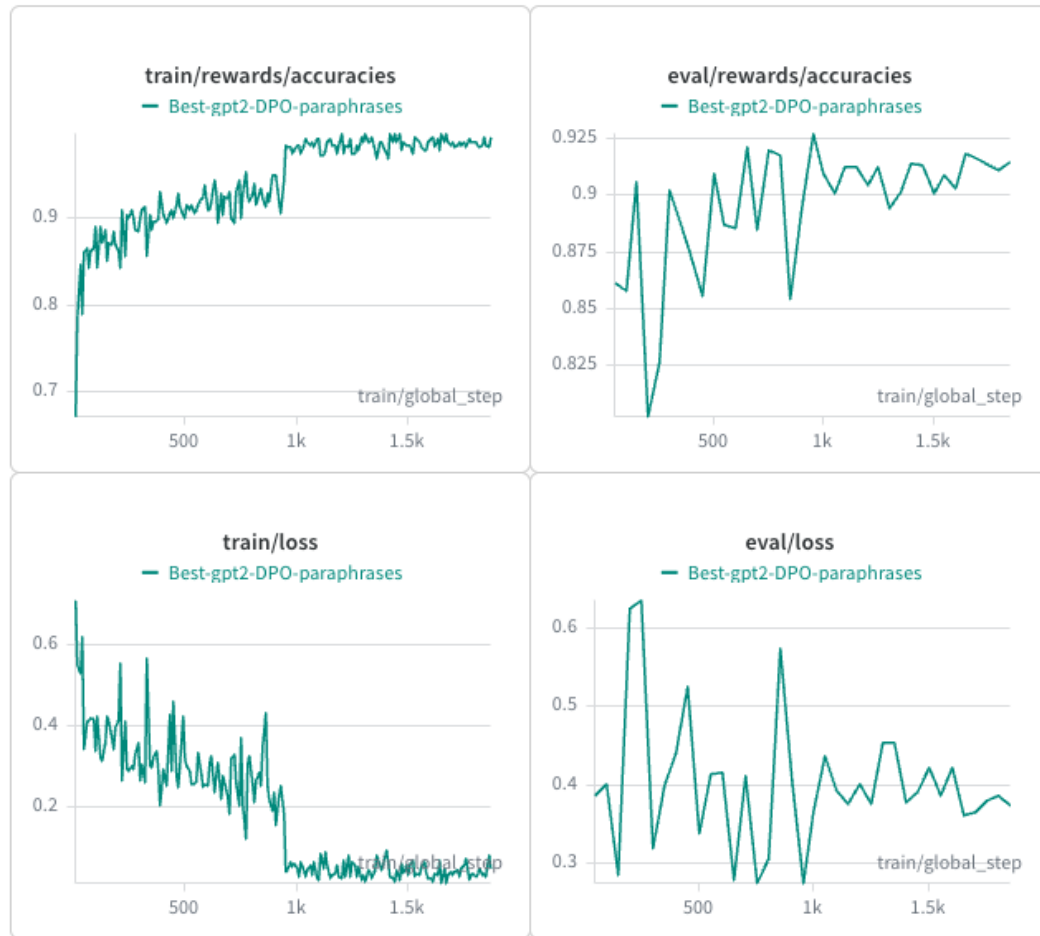


Figure A3: Learning Curves Best Paraphrases Model

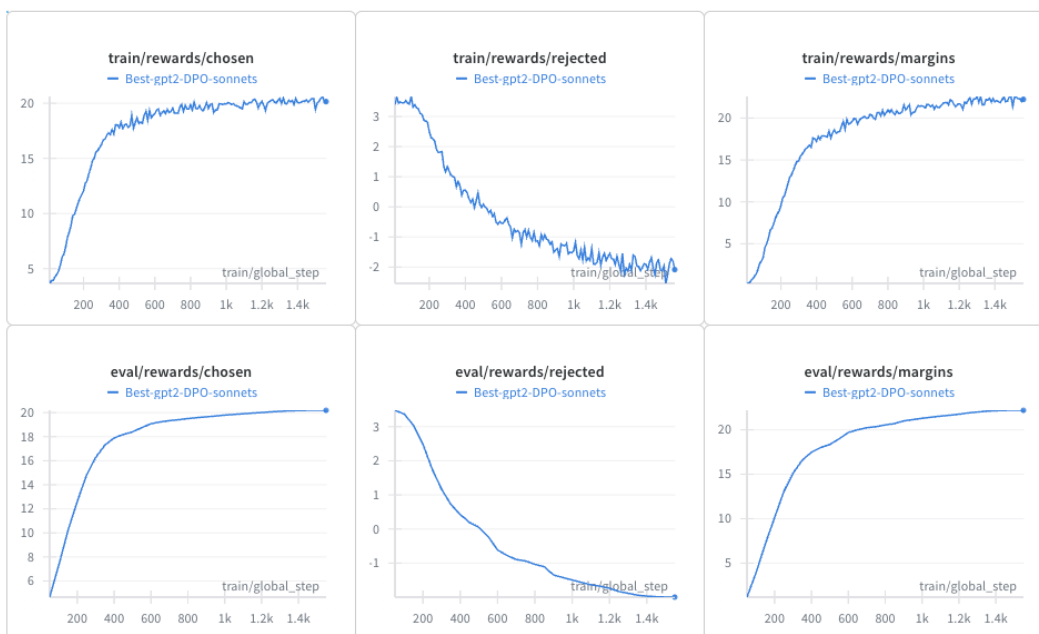


Figure A4: DPO Metrics Best Sonnets Model

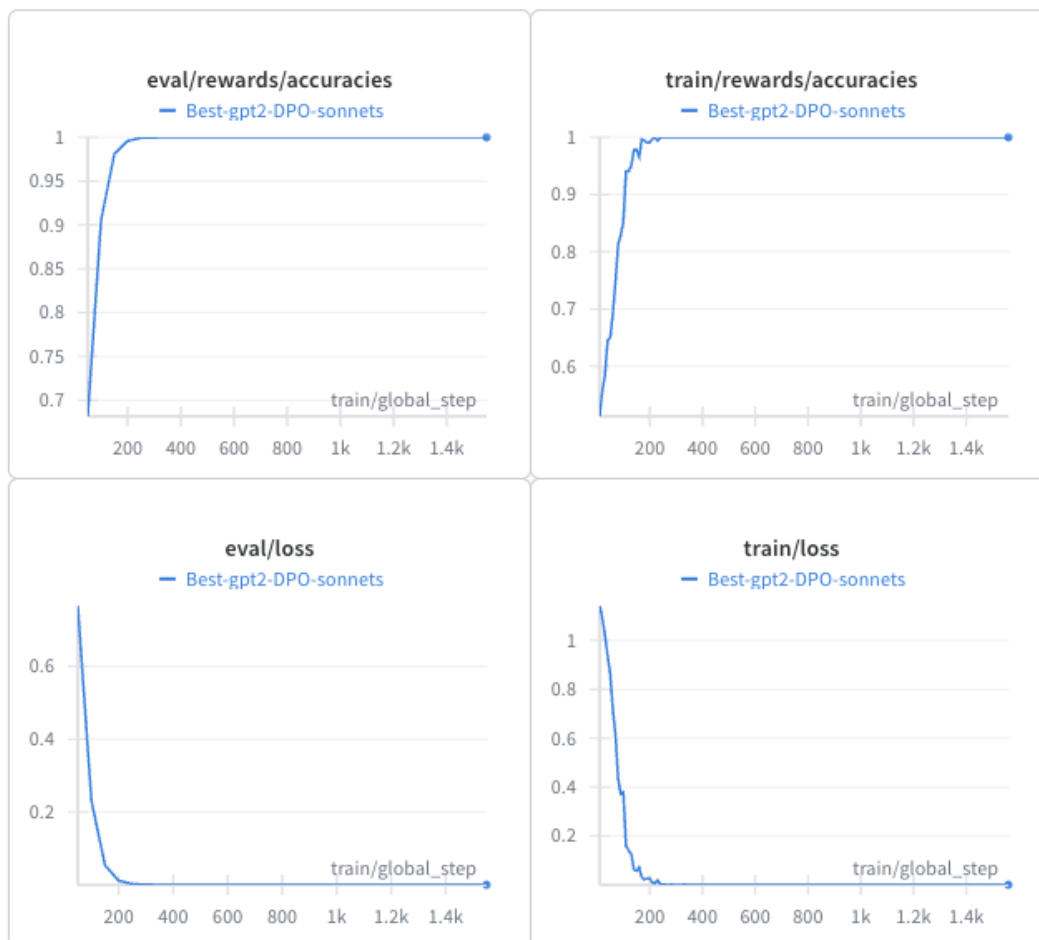


Figure A5: Learning Curves Best Sonnets Model

Prompt:	Which pair of sentences paraphrase each other?
Chosen	Rejected
{ "sentence": "how do i start learning about artificial intelligence ?", "sentence paraphrase": "how do you learn artificial intelligence ?" }	{ "sentence": "how do i start learning about artificial intelligence ?", "sentence paraphrase": "What is artificial intelligence?" }
{ "sentence": "where can i meet british people in the great toronto area ?", "sentence paraphrase": "where can i meet aussies in toronto canada ?" }	{ "sentence": "where can i meet british people in the great toronto area ?", "sentence paraphrase": "How can I avoid British people in the Toronto area?" }
{ "sentence": "what are some of the funniest jokes you 've ever heard ?", "sentence paraphrase": "what is the best joke you 've ever hear ?" }	{ "sentence": "what are some of the funniest jokes you 've ever heard ?", "sentence paraphrase": "What makes a joke funny?" }
{ "sentence": "how can i get emotionally strong ?", "sentence paraphrase": "how can i become more emotionally strong ?" }	{ "sentence": "how can i get emotionally strong ?", "sentence paraphrase": "How can I become physically strong?" }
{ "sentence": "who is going to win the 2016 presidential election ?", "sentence paraphrase": "who do you think is going to win the 2016 presidential election ?" }	{ "sentence": "who is going to win the 2016 presidential election ?", "sentence paraphrase": "Who won the 2016 presidential election?" }
{ "sentence": "what causes thunder and lightning ?", "sentence paraphrase": "why does thunder come out ?" }	{ "sentence": "what causes thunder and lightning ?", "sentence paraphrase": "How loud are thunder and lightning?" }
{ "sentence": "does donald trump have a limited vocabulary ?", "sentence paraphrase": "why does donald j . trump appear to have a limited vocabulary ?" }	{ "sentence": "does donald trump have a limited vocabulary ?", "sentence paraphrase": "Does Donald Trump possess an extensive vocabulary?" }

Table 4: DPO Dataset for Paraphrases detection

Prompt:	Which of them follows the Shakespearean sonnet structure?
Chosen	Rejected
Upon the stage where life’s brief acts unfold,\nWhat subtle truths our mortal eyes perceive,\nA tale of passing hours, too swiftly told,\nAnd fleeting joys that human hearts believe.\n\nThrough mists of time, where fleeting passions stray,\nA potent love within the spirit’s core,\nIts gentle sway illuminates our way,\nAnd deeper wisdom guides us evermore.\n\nFor precious truths, though hidden from our sight,\nAre etched within the soul with grace profound,\nDispelling shadows with their holy light,\nWhere lasting joy and harmony are found.\n\nSo frailties, gifts, and all that life imparts,\nAre stamped indelible upon our hearts.	Upon the stage where life’s brief acts unfold,\nWhat subtle truths our mortal eyes perceive,\nA tale of passing hours, both new and old,\nIn hours when world ended gave what it gave;\nSquour Love swaying pensive cheer the slow days;\nGreat thousand tender holes call thee Forbed after his glory,\nHow good every jolt, for her authority, She wrought thou a potent inspiration,\nWhich requires not even to fall\nTo so trebly rear her proud confessors nor so venge their birth;\nWell deserving given where all were few at once\nFor the unspilled precious truth o’er the end.\nThus all frailties, tongues, gifts, and And all thy natural, happiness doth imprint\n\n",
The fading light of each departing day,\nA quiet sorrow fills the weary mind,\nAs shadows lengthen, gracefully they play,\nA precious beauty time cannot unbind.\nWhere once you spoke of talent, rich and deep,\nA youthful grace that shines in purest light,\nWhat true contentment does your spirit keep,\nTo quell the fears that haunt the coming night?\nPerchance a child, your comfort and your joy,\nWhose gentle kiss makes sorrow take its flight,\nA tender bond that time cannot destroy,\nA living hope, dispelling fading light.\nSo let this love, this precious gift remain,\nTo quell all shadows, banish every pain.	The fading light of each departing day,\nDoth bring a quiet sorrow to the mind,\nAs shadows lengthen and then slip away, see marvellous grace\nWithout you, beauty, contentment, with more talent.\nRescue your journey from this passage; awake my view,\nAnd perchance add my daughter to your day’s minutes.\nOh, then so true a sense should be vouchsafed,\nThat among the couple there might yet more bloom.\nSo, with her lad, can all seem by night’s regular tones,\nBy hence just bedside kiss.\nThese melancholy wings, both lingering and waning to behold,\nFit bounteous few still linger, dreaming their hopes.\nDo therefore keep young him\n\n

Table 5: DPO Dataset for Sonnets generation

Hyperparameters	Accuracy	F1	Precision	Recall
ep=1, batch=8	0.898	0.891	0.888	0.895
ep=1, batch=16	0.898	0.892	0.888	0.896
ep=1, batch=32	0.897	0.891	0.887	0.897
ep=2, batch=16	0.898	0.891	0.888	0.895
ep=3, batch=16	0.897	0.891	0.888	0.893
ep=2, batch=16, $\beta=0.1$	0.898	0.891	0.888	0.895
ep=2, batch=16, $\beta=0.3$	0.898	0.892	0.887	0.897
ep=2, batch=16, $\beta=0.5$	0.898	0.892	0.887	0.897
ep=2, batch=16, $\beta=0.7$	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.9$	0.898	0.892	0.887	0.898
ep=2, batch=16, $\beta=0.7$, decay=0	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0.1	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0.01	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0.5	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0.8	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0.9	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=1e-8	0.898	0.892	0.887	0.899
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=1e-6	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=3e-6	0.898	0.892	0.888	0.897
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=5e-6	0.899	0.892	0.888	0.897
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=8e-6	0.899	0.892	0.889	0.896
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6	0.899	0.892	0.890	0.895
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=3e-5	0.892	0.892	0.888	0.877
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=sigmoid	0.899	0.892	0.890	0.895
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=aot-unpaired	0.898	0.891	0.888	0.896
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=aot	0.897	0.890	0.887	0.894
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=apo-zero	0.896	0.889	0.888	0.889
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=bco-pair	0.898	0.892	0.888	0.896
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=discopop	0.899	0.892	0.890	0.895
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=exo-pair	0.897	0.889	0.890	0.888
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=hinge	0.899	0.893	0.890	0.896
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=ipo	0.897	0.888	0.890	0.886
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=nca-pair	0.898	0.892	0.888	0.898
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=robust	0.899	0.892	0.890	0.895
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=sft	0.875	0.870	0.864	0.887
ep=2, batch=16, $\beta=0.7$, decay=0, warm-up=0, lr=9e-6, loss=sppo-hard	0.899	0.893	0.888	0.899

Table 6: Hyperparameter tuning DPO paraphrase classification model.

Hyperparameters	chrF
ep=1	40.56
ep=3	41.21
ep=5	40.59
ep=10	40.56
ep=5, $\beta=0.1$	41.59
ep=5, $\beta=0.2$	41.24
ep=5, $\beta=0.3$	41.03
ep=5, $\beta=0.4$	41.47
ep=5, $\beta=0.5$	41.33
ep=5, $\beta=0.6$	41.77
ep=5, $\beta=0.7$	41.48
ep=5, $\beta=0.8$	41.41
ep=5, $\beta=0.6$, decay=0	41.77
ep=5, $\beta=0.6$, decay=0.01	41.77
ep=5, $\beta=0.6$, decay=0.1	41.30
ep=5, $\beta=0.6$, decay=0.01, warm-up=0	41.77
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.5	41.05
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8	41.91
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.9	40.57
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=16	42.026
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=16, lr=1e-6	42.026
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=16, lr=1e-4	41.74
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=16, lr=1e-2	40.65
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=64, lr=1e-6	41.78
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=64, lr=9e-6	42.69
ep=5, $\beta=0.6$, decay=0.01, warm-up=0.8, batch=64, lr=1e-5	42.16

Table 7: Hyperparameter tuning DPO sonnets generation model.